JDRF tranSMART Quick Start Guide

Box Plot with ANOVA

JDRF Center of Excellence at the University of Michigan

A Box Plot with ANOVA analysis depicts how a categorical variable (e.g. Disease) differs across a continuous variable (e.g. Age).

Step 1: Select samples

Drag and drop terms of interest into the Subset boxes on the **Comparison** tab. Include all terms to be used in the analysis.

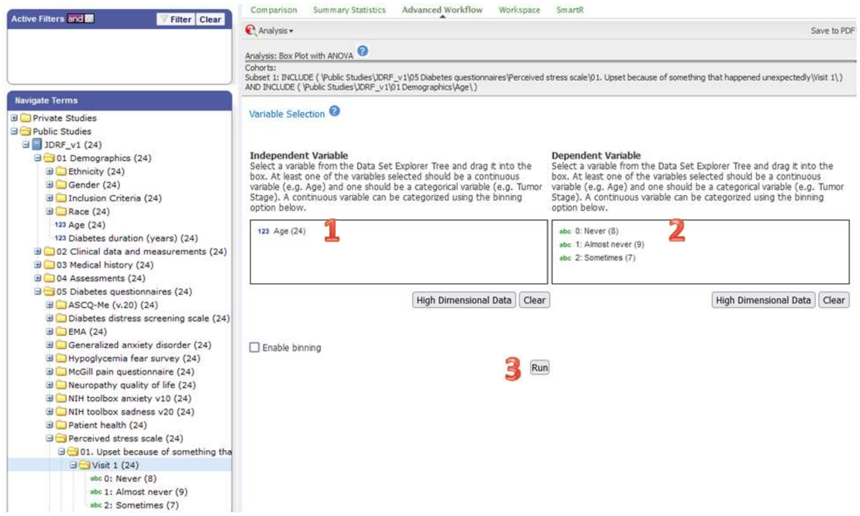
Click on the **Summary Statistics** tab to review the number of samples that will be included in the analysis. Confirm that the set of samples available is sufficient for running a meaningful analysis.

Step 2: Set up analysis

Click on the **Advanced Workflow** tab and select "Box Plot with ANOVA" from the Analysis dropdown list.

Drag terms from the Navigate Concepts tree into the Independent Variable (1) and Dependent Variable (2) boxes. You can add multiple terms, or drag over a folder to add all the terms within it. One box should contain a continuous variable (e.g. Age) and the other box should contain a categorical variable (e.g. Visit 1 answers to questions about perceived stress).

Click the "Run" button (3). A popup box will show a progress bar as the data is compiled and analyzed by tranSMART. Some analyses may take several minutes to run.



Step 3: Review results

Results will appear underneath the variable selection boxes. The Independent variable will appear along the X-axis and the Dependent variable will be the Y-axis (4).

Scroll below the box plot to see ANOVA results (5).

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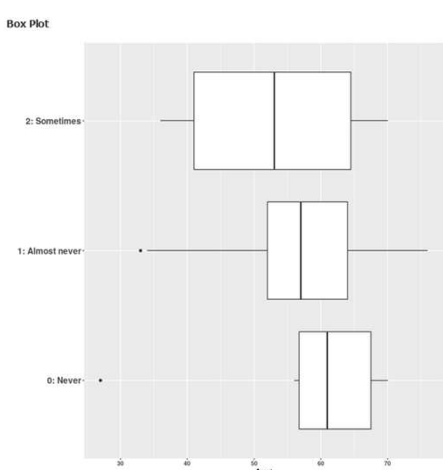

5

ANOVA Result

p-value	0.733
F value	0.316

Group	Mean	n
0: Never	58.5	8
1: Almost never	55.0	9
2: Sometimes	52.9	7

Pairwise t-Test p-Values

	0: Never	1: Almost never
1: Almost never	0.611	NA
2: Sometimes	0.444	0.764